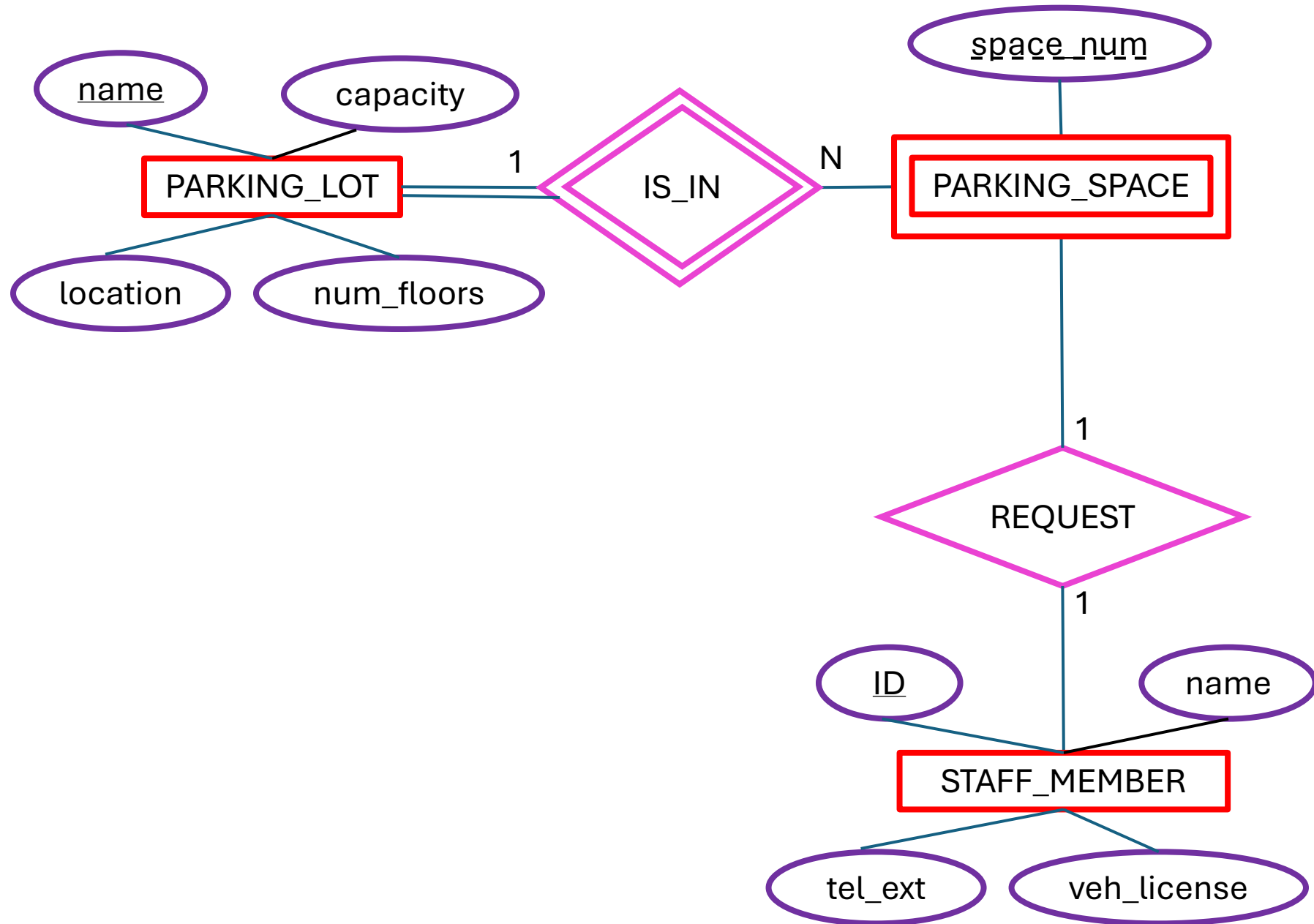


Q1



Q2

MOVIE

<u>m_id</u>	lead_a_id	d_id	...
-------------	-----------	------	-----

primary key

foreign key

ACTOR

<u>a_id</u>	p_id	d_id	...
-------------	------	------	-----

PRODUCER

<u>p_id</u>	...
-------------	-----

DIRECTOR

<u>d_id</u>	...
-------------	-----

PRODUCES

<u>p_id</u>	<u>m_id</u>
-------------	-------------

PERFORMS_IN

<u>a_id</u>	<u>m_id</u>
-------------	-------------

Q3a

EMPLOYEE			
<u>empl_id</u>	last_name	first_name	dept_id

primary key

foreign key

DEPARTMENT			
<u>dept_id</u>	dept_name	dept_location	dept_ext

SKILL	
<u>skill_id</u>	skill_desc

EMP_SKILL_SET		
<u>empl_id</u>	<u>skill_id</u>	date_qual

Q3b

```
SELECT first_name, last_name  
FROM EMPLOYEE  
WHERE dept_id= 30;
```

$$\pi_{first-name, last-name} \left(\sigma_{dept-id=30}(EMPLOYEE) \right)$$

```
SELECT first_name, last_name  
FROM EMPLOYEE E, SKILL S, EMP_SKILL_SET ES  
WHERE E.empl_id = ES. empl_id AND ES.skill_id= S. skill_id  
        AND S.skill_desc= 'C Programmer';
```

$$\pi_{first-name, last-name} \left(\sigma_{skill-desc='C Programmer'}(EMPLOYEE * EMPSKILLSET * SKILL) \right)$$

```
SELECT last_name, first_name  
FROM EMPLOYEE E, SKILL S, EMP_SKILL_SET ES  
WHERE E. empl_id = ES. empl_id AND ES. skill_id = S. skill_id  
        AND E.dept_id = 30 AND S. skill_desc = 'Java Programmer'  
ORDER BY last_name ASC;
```

$$\pi_{last-name, first-name} \left(\sigma_{skill-desc='Java Programmer' \wedge dept-id=30}(EMPLOYEE * EMPSKILLSET * SKILL) \right)$$